FINAL-YEAR PROJECT | CIE-3

TEAM RSST

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ShieldWall

An eBPF/XDP engine that stops DDoS attacks on small servers in seconds and explains itself in plain English with gpt-4o-mini.

C | Python | eBPF/XDP

Detect, Mitigate, Explain

Small servers drown in fake traffic

37%

of all DDoS attacks hit the gaming industry, the most attacked sector

$38/hr

is all a DDoS-for-hire service costs. Attacking is cheap, defending is not

94%

of web DDoS attacks stay under 100k req/s, small enough to slip past basic detection

Our focus: small operators (game servers, small VPS owners, campus networks) who can't afford Cloudflare-scale protection and have nothing automatic today.

Sources: Akamai State of the Internet, SentinelOne, StationX DDoS Statistics 2026, Cloudflare

What the system has to do

1. **Detect fast.** Spot a flood within seconds from live traffic counts.
2. **Block automatically.** Drop attack traffic in the kernel with no human in the loop.
3. **Protect real users.** Allowlist, rule cap and expiring rules so players keep playing.
4. **Explain itself.** gpt-4o-mini writes plain-English summaries but never makes blocking decisions.
5. **Run on small hardware.** One commodity server, tested in an isolated lab.

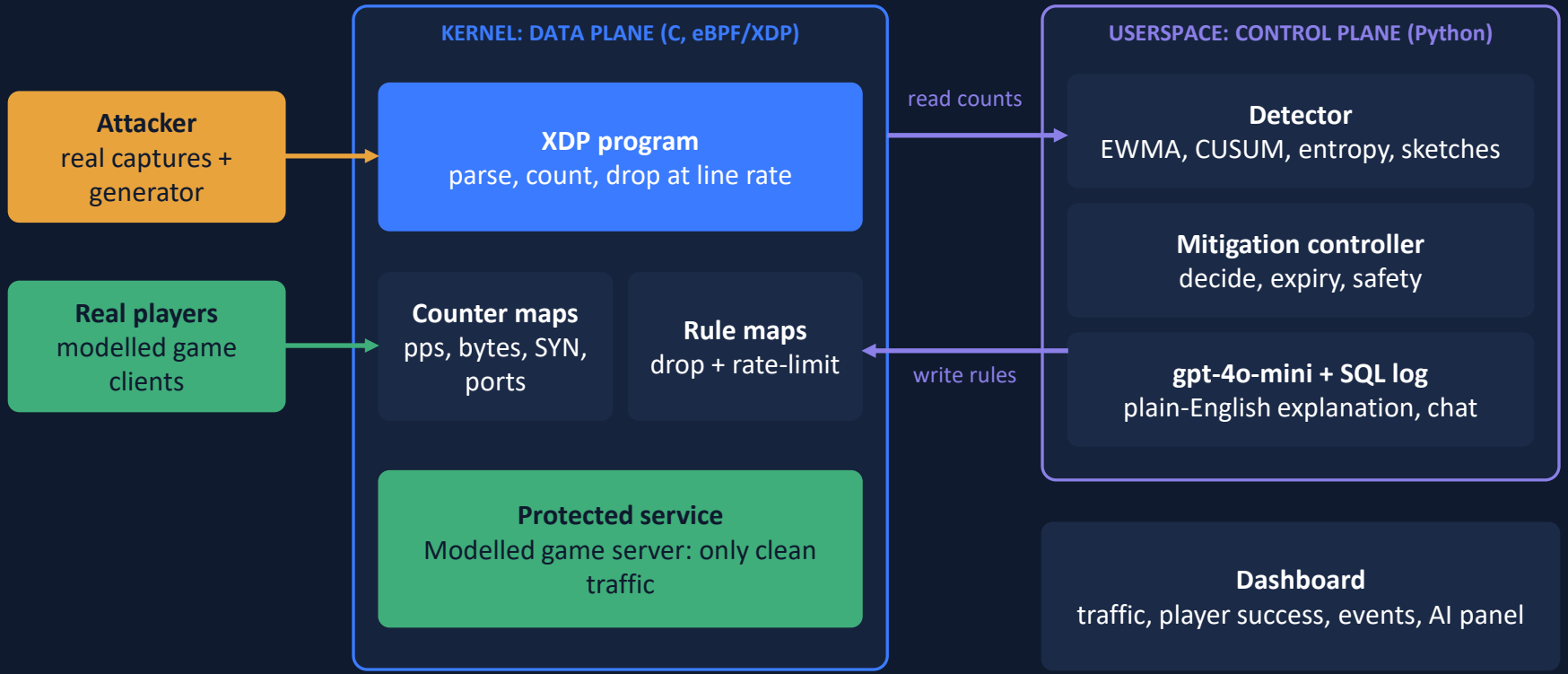
Known limit

Floods bigger than the internet link need upstream help. FlowSpec escalation is shown in our simulator, not on a real ISP.

No training needed

It learns each server's normal traffic live. No dataset, no retraining.

Design: the kernel drops, userspace decides



Implementation: detect, mitigate, explain

Detect

Three methods vote

- EWMA baseline with z-score
- CUSUM catches slow ramps
- Entropy of source spread
- Count-Min, HyperLogLog and Bloom sketches track sources in KBs
- No training: learns normal traffic live

Mitigate

Block, then undo

- Signature drop per vector
- Rate limit unknown sources
- Heavy-hitter + fragment rules
- Every rule expires in ~60 s
- Allowlist, rule cap, dry-run

Explain

Off the hot path

- gpt-4o-mini writes the incident summary
- Chat over the SQL event log
- Suggests rules; controller tests them
- Never makes a blocking decision

Stack: C + eBPF/XDP (kernel), Python + bpftool (control), SQLite, OpenAI gpt-4o-mini, scapy kernel test, live web dashboard, pytest + GitHub Actions

Evaluation and results

Real attack data

1. Public captures: 3/3 caught
2. CIC-DDoS2019: 3/3 caught
3. All blocked within 3 s
4. Kernel: 3,000/3,000 dropped

LLM check (gpt-4o-mini)

100% numbers grounded (61/61)
100% claims judged faithful
1.5 s reply, **$0.0001** each

Simulated lab	Without	With ShieldWall
Time to detect (s)	No detection	**2 s**
Time to mitigate (s)	Never	**3-4 s**
Real traffic served in attack	3-50%	**95.7-99.3%**
False alarms (93 non-attacks)	-	**0**
Detection F1 (200 trials)	-	**F1 0.97**

Real Linux kernel: 1,000/1,000 IP fragments and 2,000/2,000 packets from a flooding host dropped, every player packet passed. Table results come from our simulator.

Code and summary

GitHub repository

github.com/SushrithKbtech/shieldwall

XDP program, detection engine, dashboard, tests, kernel test and full results.

Real problem: Gaming is the top DDoS target and small operators are priced out of protection.

Real engineering: Kernel eBPF/XDP, streaming sketches, CUSUM detection and a grounded gpt-4o-mini explainer.

Real proof: real attack captures and CIC-DDoS2019 replayed and blocked; 3,000/3,000 flood packets dropped on a real Linux kernel.

A smart bouncer for small servers: it blocks attacks by itself and explains what it did.